

Wireless Intruder Alarm

JOY MUKHERJI

The wireless intruder alert system presented here incorporates a passive infra-red (PIR) movement sensor and 433MHz ready-made radio frequency (RF) transmitter and receiver modules. Should anyone break into your home, vehicle, property or warehouse, the motion sensor detector will detect the intrusion by determining changes in infra-red energy patterns and sound an alarm through the receiver unit. It has been

tested to monitor a location remotely up to hundred metres away.

Circuit and working

The circuit is divided into two sections, that is, the transmitter section and the receiver section. The transmitter section, as shown in Fig. 1, is built around PIR motion-sensor detector module HC-SR501 connected across CON2 through CON5, 433MHz transmitter module (TX1), encoder HT12E (IC1), transistors 2N2222 (T1), 2N3906 (T2) and BFG135 (T3), and a



Test Points

Test point	Details
TP0, TP3	0V (GND)
TP1, TP4	12V
TP2	Train of pulse
TP5	5V
TP6	4-5V when sensor HC-SR501 senses motion

few other components. When the PIR detects motion, its output goes high (3.3V) for three seconds. Transistors

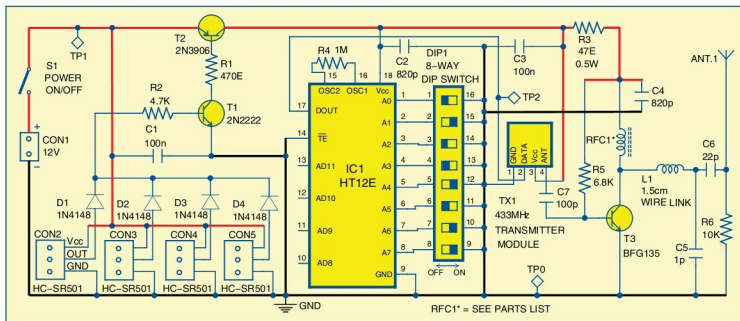


Fig. 1: Circuit diagram of the transmitter section of the wireless intruder alarm

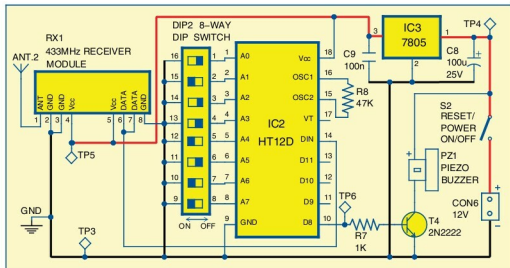


Fig. 2: Circuit diagram of the receiver section of the wireless intruder alarm

T1 and T2 conduct, and the circuit starts transmitting. Encoder IC1 generates eight address bits and four data bits, and employs amplitude shift keying (ASK) for modulation.

Data pins AD8 through AD11 of IC1 are left open (1111). Address pins A0 through A7 are set to 10110100 via DIP switch DIP1. The programmed address and data are transmitted serially via ultra-high frequency (UHF) radio-waves. The UHF frequency is better at penetrating physical barriers like walls and buildings. The address of the transmitter has to match the address of